requirements are met and that the hover control power requirements for normal operations (no failures) are easily met for all design conditions. The outlined critical failure cases are related to higher power setting flight conditions where a positive 1.05g flight path control maneuver is being commanded simultaneously with attitude control. The critical conditions are a lift-cruise fan (F5) or a front lift-fan gas generator (GG1) failure during a nose up control requirement in the pitch axis with maximum forward CG. The roll axis was not critical for any cases. Earlier analyses of the attitude control power requirements during the STOL mode had shown these requirements to be less stringent than the VTOL requirements due to the available assistance from aerodynamic controls.

TABLE 1. HOVER CONTROL ANALYSIS SUMMARY

(TOGW = 101,965 LB)

LOAD FACTOR	FAILURE CASE	ROLL CONTROL POWER REQD ~ RAD/SEC^2	CONTROL POWER AVAILABLE ~ RAD/SEC^2						PITCH CONTROL POWER REQD ~RAD/SEC^2
			ROLL		PITCH				
					FWD CG		AFT CG		
			LWU (+)	LWD (-)	NU (+)	ND (-)	NU (+)	ND (-)	
1.1	NONE	0.3	1.54	-1.54	.283	-.640	.369	-.554	.165
1.0		0.6	1.91	-1.91	.408	-.739	.486	-.661	.33
0.9		0.3	1.97	-1.97			.511	-.672	.165
1.05	GG1	0.15	0.369	-1.01	[.102]	-.268	.184	-.186	[.10]
0.90		0.15	1.194	-1.74			.388	-.396	.10
1.05	GG3	0.15	0.369	-1.013	.198	-.211	.280	-.129	.10
1.05	F1	0.15	0.577	-0.886	.149	-.240	.231	-.158	.10
0.90		0.15	1.394	-1.653			.433	-.377	.10
1.05	F3	0.15	0.378	-1.037	.142	-.280	.223	-.198	.10
1.05	F5	0.15	0.773	-0.697	[.100]	-.327	.180	-.245	[.10]